

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario-Based Assessment

DURATION: 1.5 Hrs

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario.

(5 Marks)

2. A warehouse robot uses Reinforcement Learning to find the shortest path for delivering packages while avoiding obstacles. Sometimes the robot chooses longer paths due to insufficient exploration during training. Solve how the exploration vs exploitation trade-off affects the robot's navigation performance and suggest some improvements.

(5 Marks)

3. A hospital is considering using Reinforcement Learning to optimize patient appointment scheduling in order to reduce waiting time and improve resource utilization. Demonstrate the suitability of RL for this scenario and discuss possible challenges such as delayed rewards and ethical considerations.

(5 Marks)

4. A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and suitable learning algorithm.

(5 Marks)

5. An online gaming platform uses Reinforcement Learning to match players of similar skill levels to maintain engagement. Report how reward design can influence matchmaking quality with reward definition that might lead to biased or inefficient matching results. (5 Marks)

6. A financial trading system applies Reinforcement Learning to decide buy/sell actions based on market signals. Analyze how states, actions, and rewards can be defined in this context and evaluate the risks of delayed rewards in volatile markets.

(5 Marks)

7. A drone delivery company uses RL to optimize delivery routes while avoiding restricted airspace. Construct a suitable RL framework by defining states, actions, and rewards, and justify how penalties can ensure safe navigation. (5 Marks)

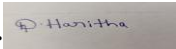
8. A university applies RL to personalize online learning paths for students. Examine how reward design can influence student engagement and defend the ethical considerations of adapting learning content dynamically. (5 Marks)

9. A traffic management authority deploys RL to control traffic lights at busy intersections. Illustrate how constraints such as emergency vehicle priority and pedestrian safety can be incorporated into the RL model and interpret the impact on congestion reduction. (5 Marks)

10. A manufacturing plant uses RL to schedule machines under resource limitations. Differentiate between traditional optimization methods and constrained RL approaches, and assess which method provides better adaptability in dynamic production environments. (5 Marks)

Code of Conduct:

*I Dasari Haritha(192425254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

SCENARIO-BASED ASSESSMENT

Q4. Smart Home Energy Management Using Reinforcement Learning

Scenario

A smart home wants to reduce electricity consumption by automatically controlling lights, fans, and air conditioners according to user presence and weather conditions.

Solution

Reinforcement Learning can be used because the smart home controller can observe the home conditions, take an appliance-control action, receive a reward, and learn the best decision over time.

1. RL Agent and Environment

In this scenario:

- **Agent:** Smart Home Controller
- **Environment:** Smart Home
- **Goal:** Reduce electricity consumption while maintaining user comfort.

The agent continuously receives information from sensors and decides how the appliances should be controlled.

2. State Space

The **state** represents the current condition of the smart home.

For example:

State = (User Presence, Temperature, Weather, Appliance Status)

Possible states:

Condition	Example
User Presence	Present / Absent
Temperature	Low / Medium / High
Weather	Cool / Normal / Hot
Light	ON / OFF
Fan	ON / OFF
AC	ON / OFF

Example scenario:

The user is **absent**, the room temperature is **high**, and the AC, fan, and light are ON.

The agent observes this as the current state.

3. Action Space

The agent controls the appliances.

Possible actions are:

A1 → Turn Light ON/OFF

A2 → Turn Fan ON/OFF

A3 → Turn AC ON/OFF

A4 → Adjust AC temperature

A5 → Keep current settings

Example:

If the user is absent:

The agent can turn OFF the light, fan, and AC to avoid unnecessary electricity consumption.

4. Reward Function

The reward tells the agent whether its decision was good or bad.

Positive reward

+10 → Electricity consumption is reduced

+5 → User comfort is maintained

Negative reward

-10 → Unnecessary electricity consumption

-15 → Appliance ON when nobody is present

-8 → User comfort is reduced

The objective is:

Maximize energy savings while maintaining user comfort.

5. Suitable RL Algorithm

Q-Learning is suitable for this scenario.

The agent initially tries different appliance-control actions. Based on the rewards it receives, it learns which action is best for each home condition.

For example:

State:

User = Absent

Temperature = High

Action:

Turn OFF AC

Reward:

+10

Result:

Q-value of this action increases.

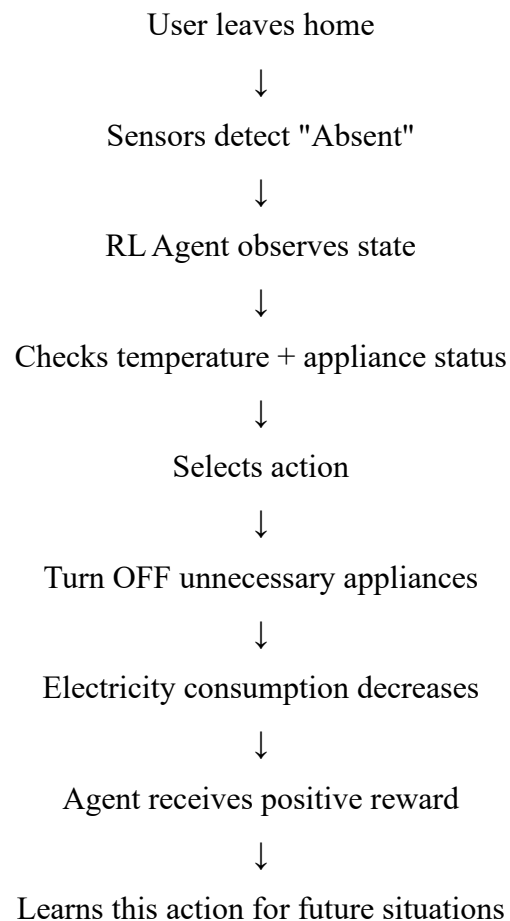
After repeated learning, the system learns to automatically switch off unnecessary appliances when the house is empty.

6. Scenario-Based Working

Consider the following situation:

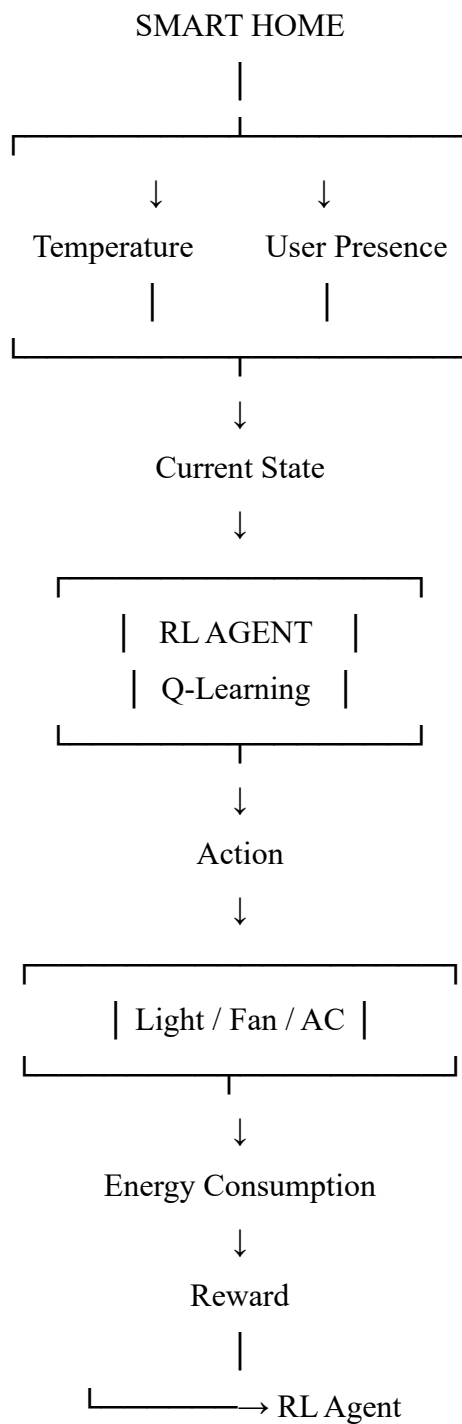
At 2 PM, the outside weather is hot and the user leaves the house. The AC, fan, and lights are still ON.

The RL system works as follows:



Later, when the user returns and the temperature is high, the agent can learn to turn the required appliances ON while maintaining comfort.

7. RL Model



8. Expected Outcome

The RL-based system should:

- **Reduce unnecessary electricity usage**
- Automatically control appliances

- Adapt to user presence
- Consider weather and temperature
- Maintain user comfort
- Improve its decisions through learning

Conclusion

In this smart-home scenario, Reinforcement Learning provides an adaptive method for controlling electrical appliances. The **state space** contains user presence, temperature, weather, and appliance status. The **action space** contains appliance-control decisions. The **reward function** encourages energy savings while maintaining comfort. A **Q-Learning agent** can learn the best control policy through repeated interaction with the smart-home environment. Thus, RL can help reduce electricity consumption without requiring fixed control rules.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand